

Beyond Assistance: The Case for Role Separation in AI-Human Radiology Workflows

Pranav Rajpurkar, PhD¹ • Eric J. Topol, MD²

In radiology, the prevailing view has long favored assistive artificial intelligence (AI) approaches, where human radiologists work “with” AI systems, under the belief that merging machine precision with human judgment naturally leads to improved diagnostic outcomes. However, recent evidence challenges this assumption by demonstrating that integrating AI into radiologists’ workflows does not always yield the anticipated enhancements. This emerging perspective suggests that a clearly defined division of labor—where AI and radiologists assume separate responsibilities in the diagnostic process—might provide more reliable results.

In the assistive AI model, AI systems provide their analysis first, followed by radiologist review and independent interpretation. The AI outputs—typically highlighting potential findings or providing quantitative measurements—are made available to radiologists as they conduct their own evaluation. This approach assumes that having access to AI insights will help radiologists catch potential oversights and make more informed decisions, while still maintaining their autonomy in the final interpretation.

However, practical implementations of the assistive model have revealed notable challenges. Agarwal et al (1) demonstrated that when discrepancies arose between AI assessment and radiologist interpretation of chest radiographs, radiologists tended to discount the AI output—even when the algorithm was correct. This *automation neglect* is only part of the story, as radiologists may have valid reasons for disregarding AI suggestions, particularly when AI systems occasionally produce clinically implausible results (2).

Conversely, radiologists can also exhibit *automation bias*—the tendency to rely overmuch on AI suggestions even when they are incorrect. In mammography interpretation, Dratsch et al (3) found that even experienced radiologists saw their diagnostic accuracy drop substantially (from 82.3% to 45.5%) when given incorrect AI predictions. This highlights that the challenges of human-AI collaboration extend beyond automation neglect to include potentially dangerous overreliance on AI systems.

These studies suggest that the integration of AI into clinical workflows faces substantial cognitive and practical challenges that must be addressed for effective implementation.

The Role Separation Framework

At its core, role separation leverages the distinct strengths of AI and radiologists by assigning them complementary yet separate responsibilities in the diagnostic workflow. Rather than merging efforts into a single, integrated process—which can lead to issues like automation neglect and automation bias—this approach divides the workflow into segments best suited to each party’s unique capabilities.

We propose three distinct models of role separation. In the *AI-first sequential model*, AI undertakes an initial portion of the workflow, handling parts of the process that can be effectively automated. Once the AI completes its portion, the radiologist takes over for expert interpretation. In the *doctor-first sequential model*, the radiologist initiates the diagnostic process using their clinical expertise, while the AI performs a distinct, complementary task that enhances the overall workflow. In the *case allocation model*, the responsibility for handling a case is determined by its characteristics. Depending on the complexity and clarity of a given case, it may be managed entirely by AI, entirely by the radiologist, or by a combination of both.

In practice, the boundaries between these approaches often blur, with many implementations involving iterative interactions rather than strict sequential processes. The implementation of AI tools in radiology workflows may need to accommodate different radiologist preferences, different institutional practices, and the specific requirements of different imaging modalities and clinical contexts.

The AI-First Sequential Model

One compelling application of the AI-first sequential model involves using AI to analyze electronic health records to prepare relevant clinical context before radiologist interpretation. In this workflow, AI systems review prior reports, laboratory results, and clinical notes to compile a focused summary of the patient’s medical history and current clinical questions relevant to the imaging study.

Bhayana et al (4) evaluated the ability of GPT-4 (OpenAI) to automatically generate clinical histories for CT requisitions from clinical notes at a cancer center. The results were encouraging: Histories generated by the large language model included more critical parameters than physician-provided requisition histories, such as primary oncologic diagnosis (99.5% vs 89%), acute symptoms (15% vs 4%), and relevant surgical history (61% vs 12%). Radiologists generally preferred the AI-generated histories for image interpretation (89% vs 5%).

However, challenges remain in implementing this approach. While AI tools can gather and structure information, maintaining context across distributed information sources remains challenging. As Johri et al (5) demonstrated, the diagnostic accuracy of large language models drops substantially when reasoning across fragmented conversation turns versus reading a consolidated patient history. This limitation applies to AI systems attempting to extract and organize relevant clinical information from fragmented electronic health record documentation, suggesting that substantial improvements in context management capabilities will be necessary before these systems can be widely deployed in clinical radiology workflows.

The Doctor-First Sequential Model

The doctor-first sequential model maintains radiologists' traditional role in primary image interpretation and diagnosis, while AI systems are deployed to enhance and extend the impact of their expertise. This model acknowledges that certain aspects of radiologic work—particularly those requiring complex pattern recognition, clinical judgment, and integration of contextual factors—remain best suited to human expertise.

Several promising applications of this model have emerged. For structured reporting, AI tools can transform a radiologist's free-text findings into standardized reports that conform to reporting guidelines. For clinical decision support, AI systems can analyze a radiologist's findings in conjunction with clinical guidelines to suggest appropriate follow-up recommendations.

Zhang et al (6) developed a large language model-based system that automatically generated impression sections from radiologist-documented findings. Radiologists first interpreted images and recorded detailed findings, then the AI system synthesized these findings into concise, structured impressions including specific diagnoses and management recommendations. This approach achieved high precision (84%) and recall (77.5%) compared with radiologist-finalized impressions, allowing radiologists to focus their expertise on image interpretation while maintaining final control over the diagnostic conclusions.

Another example is ReXplain by Luo et al (7), which allows radiologists to complete their interpretation first, then leverages AI to translate these findings into patient-friendly video reports. This approach builds on earlier work by Recht et al (8) that demonstrated that patient-centered video radiology reports substantially improved patients' understanding of their imaging results. ReXplain automates this process, making patient-centered communication more scalable while maintaining the radiologist's primary role in interpretation.

The Case Allocation Model

The case allocation model offers perhaps the most pragmatic approach to role separation in radiology, routing cases based on specific characteristics to optimize the use of both AI and human expertise. Three distinct pathways have emerged: using AI to rule out normal cases, risk-based reader allocation, and dynamic complexity-based allocation.

Using AI to Rule Out Normal Cases

This approach involves using AI to identify and clear normal cases, while routing all other cases to radiologists for interpretation. ChestLink (Oxipit) received Conformité Européenne marking in 2022, permitting fully autonomous reporting of normal chest radiographs throughout the European Economic Area—an important milestone. Given that a large proportion of screening examinations are normal, this approach can substantially reduce radiologist workload while maintaining safety. Similar successes have been seen in ophthalmology with the Food and Drug Administration–cleared AEYE-DS tool (AEYE Health) and in dermatology with the DERM system (Skin Analytics), which achieved a 99.8% negative predictive value for ruling out melanoma—surpassing the 98.9% rate for dermatologists—ensuring safe deferral of clearly benign lesions.

Risk-based Reader Allocation

This approach involves using AI to triage cases between single- and double-reading pathways based on risk assessment. The MASAI trial by Hernström et al (9) demonstrated the effectiveness of this approach. In this randomized controlled trial of over 100 000 women, AI was used to triage screening examinations: Cases with low to intermediate risk received single reading, while high-risk cases underwent double reading. This risk-based allocation led to a 29% increase in cancer detection compared with standard double reading (6.4 vs 5.0 cancers per 1000 individuals screened) while reducing radiologist workload by 44%. Importantly, the increased detection was primarily of small, lymph node–negative invasive cancers, suggesting the potential for earlier diagnosis of clinically relevant disease.

Dynamic Complexity-based Allocation

This approach involves dynamically allocating cases based on case complexity and AI uncertainty levels. Agarwal et al (1) found that while access to AI predictions improved radiologist performance for cases where the AI predictions were very close to 0 or 1 (indicating high confidence in either direction), the performance of the AI system itself was even better than the AI-assisted radiologists in these ranges.

The PRAIM study (10) found that radiologists naturally gravitated toward efficient role separation when given flexibility in workflow design. They tended to use AI-supported viewing for cases the AI system tagged as “normal,” while preferring traditional viewing for more complex cases. This natural separation emerged without explicit guidance, suggesting that radiologists intuitively recognize when to leverage AI's strengths versus maintain direct control.

Dvijotham et al (11) advanced this concept with their Complementarity-Driven Deferral to Clinical Workflow (CoDoC) system, which learns to decide between the opinion of a predictive AI model and a clinical workflow. In breast cancer screening, CoDoC reduced false positives by 25% (at the same false-negative rate) compared with double reading with arbitration, while achieving a 66% reduction in clinician workload.

The implementation of these allocation models requires careful consideration of liability, regulatory frameworks, and payment models. Saenz et al (12) noted that with truly autonomous AI systems, liability may shift substantially from physicians to the AI companies, which are adapting by purchasing insurance and negotiating contractual safeguards.

Navigating the Path Forward

The emergence of more powerful “generalist medical AI” systems (13) may help overcome previous limitations and enable more effective assistance in clinical workflows. MedVersa (14) demonstrates this potential, offering a single system capable of examining different types of scans, drafting reports, detecting diseases, and measuring anatomic structures. Rajpurkar and Topol (15) proposed addressing integration challenges through a clinical certification pathway that mirrors medical training, with systems progressing through stages of increasing responsibility while remaining under rigorous performance monitoring.

The radiology community faces difficult decisions about how to proceed. Current evidence suggests that neither fully integrated assistive approaches nor complete automation represent

optimal solutions. Instead, a careful, measured approach to role separation—guided by rigorous clinical validation and real-world evidence—offers the most pragmatic path forward. Successful implementation will likely require flexibility to accommodate different practice patterns, radiologist preferences, and institutional contexts, recognizing that the most effective workflows may blend elements from multiple models rather than adhering rigidly to any single approach.

Author affiliations:

¹ Department of Biomedical Informatics, Harvard Medical School, Boston, Mass

² Scripps Research Translational Institute, 3344 N Torrey Pines Ct, Ste 300, La Jolla, CA 92037

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Address correspondence to: E.J.T. (email: etopol@scripps.edu).

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